Ethical Considerations

A main ethical consideration in my capstone project is making sure that the data is consistent and accurate. There are two components to collecting shot data from the game film that each have their own ethical considerations: the Sportscode data and the shot characteristics data. The Sportscode data captures the tagged timeline of each game. It records the opponent, start time of the shot, the duration, number of shots in a game, the type of possession (counter attack, 6on5, 7on5, penalty, or even), and the result of the shot (goal or miss). This data is entered by a player or coach either during or after the game, which leaves room for human error in the tagging process. The film-tagger may accidentally give a shot to the wrong team, start the shot tag way before the shot took place, or accidentally exclude the type of possession of the shot. While these errors are a concern in the accuracy of the shot data, they are not a large concern of this project. The main concentration is creating a shot-analysis dashboard with the tagged data the PPWWP team already has. The shot characteristics data is manually-entered observations that I will enter in a coded Excel sheet with dropdown menus to avoid typos and misspelled data. The shot characteristics capture the shot type, the shooter's location, if the shooter was a center, the shot location in the goal, if there was defensive pressure, and the shot-clock time. This data will be entered as I watch the tagged game footage, which also leaves room for human error. Some shot characteristic entries may be subjective. For example, if a shooter is in between two positions, I may record a position that another reviewer disagrees with. While human error and bias cannot be completely avoided, I will carefully define each manual shot characteristic in a data dictionary to address human error the best I can.

Another ethical consideration is the potential for bias in my selection of games and shot characteristics. I am focusing on games played against the nationally ranked top 25 opponents and Claremont-Mudd-Scripps rather than using all the games played during the 2026 season. While this keeps games competitive and makes the amount of film manageable, it additionally means the dataset may not fully represent PPWWP's overall shooting performance, as PPWWP was ranked just outside the top 25 teams for the 2026 season. Many games against top-ranked opponents resulted in losses, and the shooting statistics could reflect the opponent's difficulty rather than the PPWWP team's typical performance. The selection of shot characteristics may also create some ethical considerations. The manually coded shot-characteristics may not capture every aspect of a shot, due to the nuances of sports games, however, they will capture most. To address these biases the best I can, I will create filters such as a score-differential slider, opponent selection, and shot characteristic selection, which will allow coaches to filter shot statistics in the context of how competitive matches were. This won't eliminate bias, but will help viewers interpret the results.

Finally, because the project uses team-owned footage and individual player statistics, a large ethical consideration is player privacy and how the data is shared. I plan to limit the identifiable information of players and work with the coaching staff to determine how the dashboard will be used. Ultimately, my goal is to create a system that collects data and presents it in a useful, accurate, and reliable way. While I have not yet completed the full coding process or application, I am incorporating these considerations into the project's design.